Animesh Biswas

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EDUCATION

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar, Odisha
B. Tech, Electronics and Telecommunication Engineering
CGPA: 8.41/10.0 | Expected Graduation: May 2027

Jamshedpur Public School, Jamshedpur, Jharkhand
Higher Secondary Certificate (CBSE Class XII) | Completed: March 2023

RESEARCH EXPERIENCE

16-Bit DSP–CMOS Pipelined ALU Design | *KIIT University (Dec 2026)*
Tools: Xilinx Vivado Design Suite, Verilog HDL, Digilent Nexys 4 DDR (Xilinx Artix-7)

- Developed a 6-stage pipelined architecture on Nexys 4 DDR, optimizing throughput by explicitly inferring DSP48E1 hard macros for multiplication.
- Achieved 100 MHz operational frequency and low dynamic power (25mW) by implementing latency balancing and register retiming strategies.

Development of Proportionate-Type & VLSI Algorithms | Minor Research (Ongoing)
Tools: MATLAB, Verilog HDL, Xilinx Vivado, FPGA

- Simulating proportionate-type adaptive algorithms for system identification to analyze convergence and performance baselines in MATLAB.
- Developing the VLSI architecture by translating high-level MATLAB models into synthesizable Verilog HDL for hardware acceleration on FPGA.

PROJECTS

Autonomous Sand Rover V2 | *Kshitij, IIT Kharagpur (Jan 2025)*
Tools: Autodesk Fusion 360, Arduino (C), ESP32, ESCs

- Re-engineered the drive system using Electronic Speed Controllers (ESCs) and custom firmware to achieve smoother, precise manoeuvrability.
- Integrated a camera module for low-light navigation and upgraded the mechanical chassis with improved traction for autonomous cave exploration.

Autonomous Sand Rover | *Kshitij, IIT Kharagpur (Jan 2024)*
Tools: Autodesk Fusion 360, Arduino (C), Motor Drivers (L293D)

- Engineered the rover's 3D mechanical model and chassis using Autodesk Fusion 360, optimizing geometry for stability in sand terrain.
- Implemented motor control logic using L293D drivers and Arduino C to enable basic navigation and mobility functions.

RF Mobile Phone Detector (Design Phase)
Tools: Raspberry Pi Pico, Micro Python/C, MX-05 RF Module, Circuit Simulation

- Architected a low-cost detector using Raspberry Pi Pico to identify 433 MHz RF bursts via custom threshold-based logic.
- Finalized the complete circuit schematic and scalable system design to support future wideband signal detection upgrades.

TECHNICAL SKILLS

Languages: C, Python, MATLAB, Verilog
Software: KiCad, Tina-Ti, Xilinx Vivado, VS Code, Arduino IDE
Hardware: Microcontrollers (AVR, ESP32, Raspberry Pi Zero), FPGA, PCB Design & Prototyping
Domains: Analog & Digital Circuit, Power Electronics, Semiconductor Devices